43. Load Balancing and Reverse Proxy (Linux view)

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Phase: Phase 6 — Networking
Estimated time: 1–2 days
Slug: `43-http-load-tls`

Learning objectives

- Explain reverse proxy role
- Read nginx/Caddy basics at config level
- Terminate TLS at proxy
- Health checks and upstreams

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics.

Linux hosts often run nginx/HAProxy/Envoy as edge or sidecars.

Concepts:

- Upstream pool
- Health checks
- TLS termination vs passthrough
- Access/error logs

You will configure these repeatedly in AI/gateway and microservice designs.

Common mistakes

- Single upstream without health checks
- Weak TLS configs
- Logging secrets at info level

Exercises

Work in `exercises/43-http-load-tls/`. Run:

```
./test.sh
```

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** note one real command you ran on a Linux host related to this lesson
- **C (Challenge):** rewrite your solution from memory the next day

Do not open `solutions/43-http-load-tls/` until you have an honest failing attempt.

Self-check

1. What is TLS termination?
2. Why health checks?
3. Where find nginx error log?

Further reading

- https://nginx.org/en/docs/